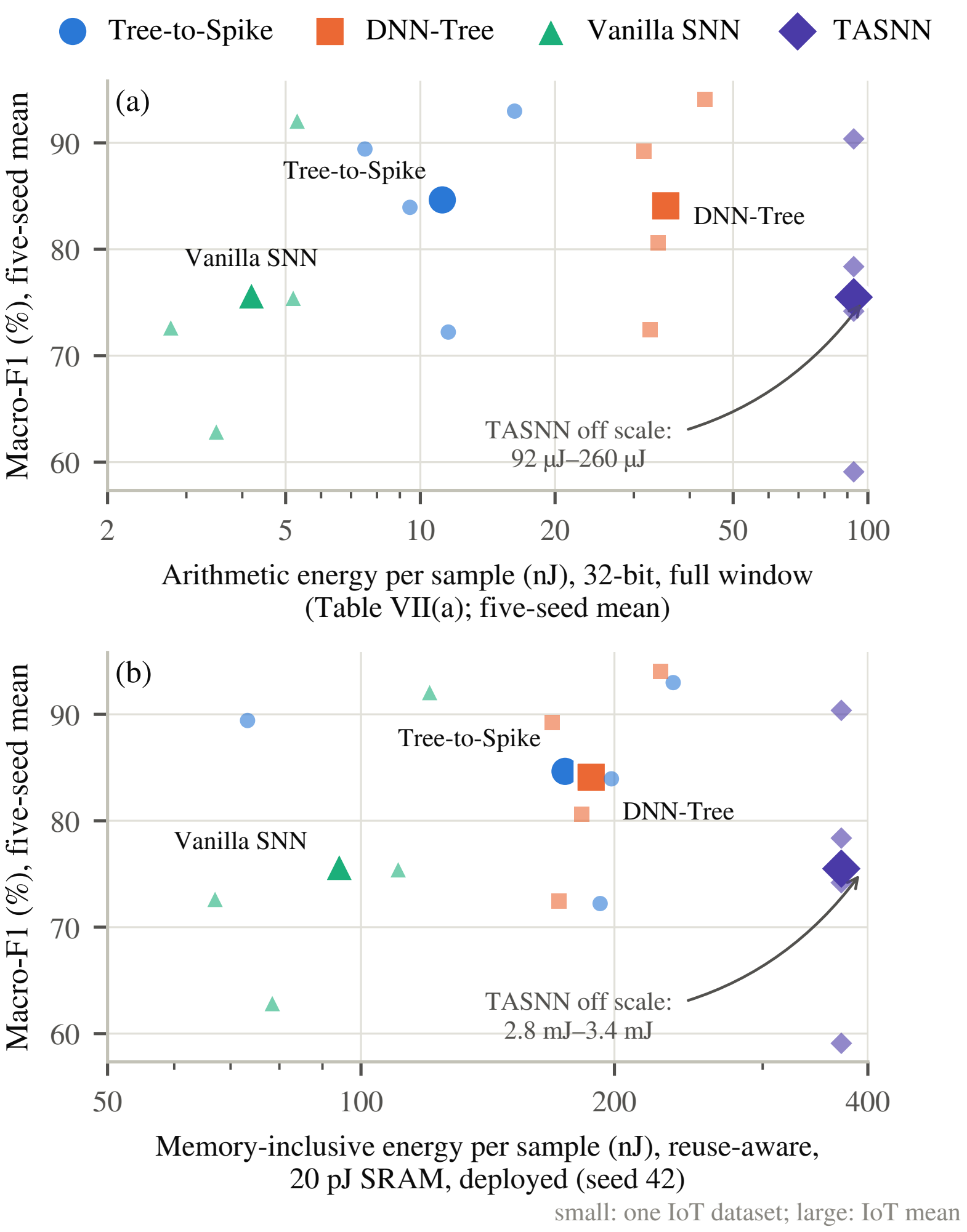
Tree-to-Spike
DNN-Tree
Vanilla SNN
TASNN
(a)
Macro-F1 (%), five-seed mean
90
80
70
60
Tree-to-Spike
DNN-Tree
Vanilla SNN
TASNN off scale:
92 µJ–260 µJ
2
5
10
20
50
100
Arithmetic energy per sample (nJ), 32-bit, full window
(Table VII(a); five-seed mean)
(b)
Macro-F1 (%), five-seed mean
90
80
70
60
Tree-to-Spike
DNN-Tree
Vanilla SNN
TASNN off scale:
2.8 mJ–3.4 mJ
50
100
200
400
Memory-inclusive energy per sample (nJ), reuse-aware,
20 pJ SRAM, deployed (seed 42)
small: one IoT dataset; large: IoT mean